

NON-STEADY GREEN'S FUNCTIONS FOR AN INFINITE STRIP OF LEAKY AQUIFER

M. S. Hantush and C. E. Jacob

Abstract--Solutions are obtained for the non-steady flow to a well in an infinite-strip of leaky aquifer whose boundaries are maintained either at a constant head or at a vanishing flux. These solutions are used to deduce the corresponding solutions for non-leaky aquifers. Also they are used to obtain the non-steady Green's functions for the infinite-strips. Green's functions are tabulated for infinite quadrants and half-planes.

Introduction--The partial differential equation for non-steady flow in leaky aquifers has been integrated for special cases [JACOB, 1946; HANTUSH, 1949; HANTUSH and JACOB, 1954]. The equation in polar coordinates is

$$(\partial^2 s / \partial r^2) + (1/r) (\partial s / \partial r) - (s/B^2) = (S/T) (\partial s / \partial t) \quad \dots \dots \dots (1)$$

where s = the drawdown at a point a distance $r = \sqrt{(x - x_1)^2 + (y - y_1)^2}$ from the center of the well (x_1, y_1) and at time t from the start; S = storage coefficient (volume of water which a unit decline of head releases from storage in a vertical prism of the aquifer of unit cross section); T = transmissibility of the aquifer (discharge per unit normal width per unit hydraulic decline); B = leakage factor, which is equal to $(Tb'/K')^{1/2}$ [HANTUSH and JACOB, 1954], b' and K' are respectively the thickness and the transmission constant or 'permeability' of the semi-impervious confining bed, through which vertical leakage takes place in proportion to the drawdown.

In the present paper it is shown how the solution obtained in the paper on Non-steady radial flow in an infinite leaky aquifer [HANTUSH and JACOB, 1955] can be utilized to obtain a solution for the non-steady flow in an infinite-strip of a leaky aquifer, if the boundaries of the strip are kept either at a constant head or at a vanishing flux.

The systems correspond to a well located between two parallel rivers or canals, or, in the case where there is no flux across the boundaries, to a well between parallel impermeable boundaries, such as two parallel faults or buried bedrock walls or ledges. The solutions are of practical interest, especially in artificial drainage of land and in relieving pressures underneath river levees.

Analysis--In this problem of a leaky aquifer, a body of ponded water or a bed of sand in which there is a water table overlies a semi-pervious bed of thickness b' and permeability K' . The latter confines an artesian sand of thickness b and permeability K , which rests on an impervious layer. A well is screened in the artesian sand.

The drawdown produced by a well starting from rest and discharging at a steady rate from such an aquifer of uniform thickness and permeability and infinite extent [HANTUSH and JACOB, 1955] is given by

$$s/(Q/4\pi T) = \int_u^\infty \exp[-x - (r^2/4B^2)/x] (dx/x) \quad \dots \dots \dots (2)$$

where Q = discharge of well, $u = (r^2 S/4Tt)$, the remaining symbols being as defined above.

Making the substitution $x = r^2 Z$, (2) becomes

$$s/(Q/4\pi T) = \int_{(S/4Tt)}^\infty \left\{ \exp[-r^2 Z - (1/4B^2 Z)] \right\} (dZ/Z) \quad \dots \dots \dots (3)$$

Due to the linearity and homogeneity of (1), the method of images can be used to obtain zero drawdown or vanishing flux or a combination of both on the boundaries of different geometrical forms.

The appropriate image system which results in zero drawdown along the boundaries of the infinite-strip; $2a$ wide, is shown in Figure 1.

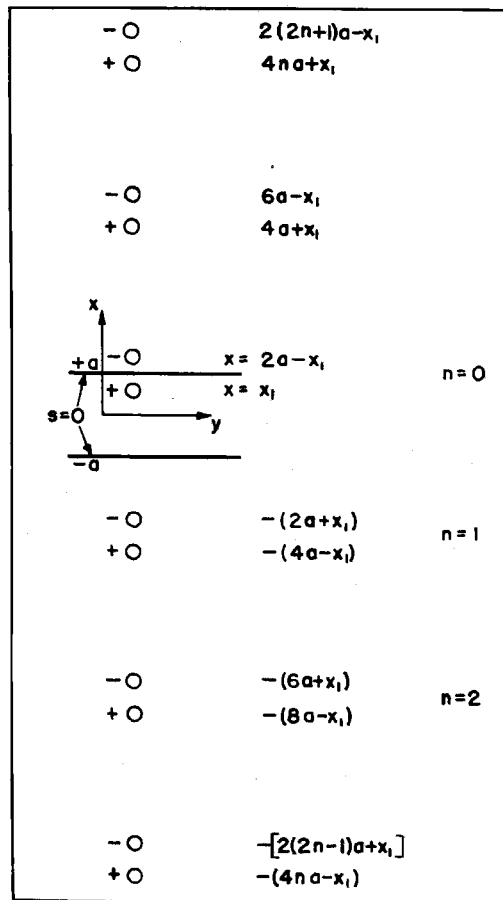


Fig. 1--Image system which gives zero drawdown along $x = \pm a$; the actual well is at (x_1, y_1) ; the images extend to $\pm\infty$; discharging wells and recharge wells are indicated by (+) and (-) signs respectively

The drawdown distribution in the infinite-strip is the resultant of the drawdown produced by all the images of this system and may, therefore, be written as

$$s/(Q/4\pi T) = \sum_{n=-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{(S/4Tt)}{(S/4Tt)} \left(\exp \left\{ -Z [(x-x_1-4na)^2 + (y-y_1)^2] \right\} \right. \\ \left. - \exp \left\{ -Z [(x+x_1-2a-4na)^2 + (y-y_1)^2] \right\} \right) [\exp(-1/4B^2Z)] (dZ/Z) \dots (4)$$

Interchanging summation and integration [BROMWICH, 1947] and simplifying, (4) becomes

$$s/(Q/4\pi T) = \int_{(S/4Tt)}^{\infty} \left\{ \exp [-(y-y_1)^2 Z - 1/4 B^2 Z] \right\} (\Sigma) (dZ/Z) \dots \dots \dots (5)$$

where

$$\Sigma = \sum_{n=-\infty}^{\infty} \left(\exp \left\{ -16a^2 Z [(x-x_1)(1/4a) - n]^2 \right\} - \exp \left\{ -16a^2 Z [(x+x_1-2a)(1/4a) - n]^2 \right\} \right) \dots \dots \dots (6)$$

Eq. (6) can be reduced as follows. First, by the aid of the integral relation [INGERSOLL and ZOBEL, 1948]

$$\sqrt{2/\pi} \int_0^{\infty} [\exp(-p^2 t^2)] \cos(\alpha t) dt = (1/\sqrt{2p^2}) \exp(-\alpha^2/4p^2)$$

the following relation can readily be obtained

$$\sqrt{2/\pi} \int_0^{\infty} \left\{ [\exp(-p^2 t^2)] \cos(bt) \right\} \cos(\alpha t) dt = (1/2 \sqrt{2p^2}) \left\{ \exp[-(\alpha+b)^2/4p^2] + \exp[-(\alpha-b)^2/4p^2] \right\} \dots \dots \dots (7)$$

Also, from (7), the Fourier cosine transform for the function

$$f(t) = [\exp(-p^2 t^2)] \cos(bt) \dots \dots \dots (8)$$

is given by

$$F_c(\alpha) = (1/2 \sqrt{2p^2}) \left\{ \exp[-(\alpha+b)^2/4p^2] + \exp[-(\alpha-b)^2/4p^2] \right\} \dots \dots \dots (9)$$

Poisson's formula [TITCHMASH, 1937] gives the following relation

$$\sqrt{\beta} \left[(1/2) F_c(0) + \sum_{n=1}^{\infty} F_c(n\beta) \right] = \sqrt{\theta} \left[(1/2) f(0) + \sum_{n=1}^{\infty} f(n\theta) \right] \dots \dots \dots (10)$$

where $f(t)$ = a function that tends to zero as $t \rightarrow \infty$; $F_c(\alpha)$ = the Fourier cosine transform of $f(t)$; and $\theta\beta = 2\pi$ with $\theta > 0$.

Taking $\beta = 1$ and $\theta = 2\pi$ and substituting in (10) the equivalent expressions of $f(t)$ and $F_c(\alpha)$ from the relations (8) and (9), one obtains

$$\begin{aligned} (1/2 \sqrt{2p^2}) \left(\exp(-b^2/4p^2) + \sum_{n=1}^{\infty} \left\{ \exp[-(n+b)^2/4p^2] + \exp[-(n-b)^2/4p^2] \right\} \right) \\ = \sqrt{2\pi} \left(1/2 + \sum_{n=1}^{\infty} \left\{ \exp[-(2n\pi p)^2] \right\} \cos(2n\pi b) \right) \dots \dots \dots (11) \end{aligned}$$

Letting n in the first sum of the left-hand member of (11) take negative values, and combining the result, one obtains

$$\sum_{n=-\infty}^{\infty} \exp[-(b-n)^2/4p^2] = 2 \sqrt{4\pi p^2} \left(1/2 + \sum_{n=1}^{\infty} \left\{ \exp[-(2n\pi p)^2] \right\} \cos(2n\pi b) \right) \dots (12)$$

Taking $b = (x - x_1)/4a$ and $(1/4p^2) = 16a^2Z$, (12) reduces to

$$\begin{aligned} \sum_{n=-\infty}^{\infty} \exp \left\{ -16a^2Z \left[(x - x_1)(1/4a) - n \right]^2 \right\} &= (1/2a) \sqrt{\pi/Z} \left(1/2 \right. \\ &+ \sum_{n=1}^{\infty} \left\{ \exp \left[- (n\pi/2a)^2/4Z \right] \right\} \cos \left[n\pi (x - x_1)/2a \right] \left. \right\} \dots \dots \dots (13) \end{aligned}$$

Again, taking $b = (x + x_1 - 2a)(1/4a)$ and $(1/4p^2) = 16a^2Z$, (12) gives

$$\begin{aligned} \sum_{n=-\infty}^{\infty} \exp \left\{ -16a^2Z \left[(x + x_1 - 2a)(1/4a) - n \right]^2 \right\} &= (1/2a) \sqrt{\pi/Z} \left(1/2 \right. \\ &+ \sum_{n=1}^{\infty} \left\{ \exp \left[- (n\pi/2a)^2/4Z \right] \right\} \cos \left[n\pi (x + x_1 - 2a)/2a \right] \left. \right\} \dots \dots \dots (14) \end{aligned}$$

Subtracting (14) from (13) and simplifying, one finds

$$\Sigma = (1/a) \sqrt{\pi/Z} \sum_{n=1}^{\infty} R_n \exp \left[- (n\pi/2a)^2/4Z \right] \dots \dots \dots (15)$$

where Σ is the sum in (6) above, and where

$$R_n = \sin \left[n\pi (x - a)/2a \right] \sin \left[n\pi (x_1 - a)/2a \right] \dots \dots \dots (16)$$

On replacing Σ in (5) by its equivalent from (15) and reversing the order of summation and integration, (5) can be expressed as

$$s/(Q/4\pi T) = \sum_{n=1}^{\infty} R_n \int_{(S/4Tt)}^{\infty} (\sqrt{\pi/a}) \left\{ \exp \left\{ - (y - y_1)^2 Z - [1/B^2 + (n\pi/2a)^2]/4Z \right\} \right\} (dZ/Z^{3/2}) \dots \dots \dots (17)$$

The infinite integral in (17), designated by I , is evaluated as follows

$$\begin{aligned} I &= (\sqrt{\pi/a}) \int_{(S/4Tt)}^{\infty} \left\{ \exp \left\{ - (y - y_1)^2 Z - (\pi/2a)^2 [n^2 + (2a/\pi B)^2]/4Z \right\} \right\} (dZ/Z^{3/2}) \\ &= (8/\sqrt{\pi}) \left[1/\sqrt{n^2 + (2a/\pi B)^2} \right] \int_0^{\lambda_1} \exp \left\{ - \lambda^2 - (\pi/4a)^2 [n^2 + (2a/\pi B)^2] (y - y_1)^2 (1/\lambda^2) \right\} d\lambda \dots \dots \dots (18) \end{aligned}$$

where the following substitution has been made

$$\lambda = [\pi \sqrt{n^2 + (2a/\pi B)^2}] / (4a \sqrt{Z})$$

also, λ_1 stands for $[\pi \sqrt{n^2 + (2a/\pi B)^2}] / (4 \sqrt{a^2 S/4Tt})$.

Eq. (18) can be changed to a more useful form with the aid of the integration formula [BINET, 1841]

$$\begin{aligned} (2/\sqrt{\pi}) \int_0^{\alpha_1} \exp \left[-\alpha^2 - (\beta/\alpha)^2 \right] d\alpha &= (1/2) \left[\exp (-2\beta) \right] \\ &\times \operatorname{erfc} \left[(\beta/\alpha_1) - \alpha_1 \right] - (1/2) \left[\exp (2\beta) \right] \operatorname{erfc} \left[(\beta/\alpha_1) + \alpha_1 \right] \end{aligned}$$

where the complementary error function is

$$\operatorname{erfc}(x) = 1 - \operatorname{erf}(x) = (2/\sqrt{\pi}) \int_x^{\infty} e^{-\gamma^2} d\gamma$$

Eq. (18) may, therefore, take the form

$$I = (2/\delta) \left(\left\{ \exp [-(\pi \delta/2a) |y - y_1|] \right\} \operatorname{erfc} \left\{ \sqrt{(y - y_1)^2 S/4Tt} - [(\pi \delta/4)/\sqrt{a^2 S/4Tt}] \right\} \right. \\ \left. - \left\{ \exp [(\pi \delta/2a) |y - y_1|] \right\} \operatorname{erfc} \left\{ \sqrt{(y - y_1)^2 S/4Tt} + [(\pi \delta/4)/\sqrt{a^2 S/4Tt}] \right\} \right) \dots \dots \dots (19)$$

where $\delta = \sqrt{n^2 + (2a/\pi B)^2}$.

Drawdown distribution for zero drawdown on the boundaries in a leaky aquifer--Finally, on inserting (19) in (17), the drawdown distribution will be

$$s/(Q/2\pi T) = \sum_{n=1}^{\infty} (R_n/\delta) \left(\left\{ \exp [-(\pi \delta/2a) |y - y_1|] \right\} \operatorname{erfc} \left\{ \sqrt{u_y} - [(\pi \delta/4)/\sqrt{u_a}] \right\} \right. \\ \left. - \left\{ \exp [(\pi \delta/2a) |y - y_1|] \right\} \operatorname{erfc} \left\{ \sqrt{u_y} + [(\pi \delta/4)/\sqrt{u_a}] \right\} \right) \dots \dots \dots (20)$$

where $R_n = \sin [n\pi(x - a)/2a] \sin [n\pi(x_1 - a)/2a]$; $u_y = (y - y_1)^2 S/4Tt$; and $u_a = a^2 S/4Tt$.

Since the values of the terms of the series of (20) can be found from tables and since the series tends to zero quite rapidly as n increases, it can be seen that the equation is not an inconvenient one for the computation of the drawdown.

Recalling that $\operatorname{erfc}(-\infty) = 2$ and $\operatorname{erfc}(\infty) = 0$, as time becomes great, that is, as $t \rightarrow \infty$, (20) is seen to take the form

$$s/(Q/2\pi T) = 2 \sum_{n=1}^{\infty} (R_n/\delta) \exp [-(\pi \delta/2a) |y - y_1|] \dots \dots \dots (21)$$

which as should be expected is the steady state solution of the problem [HANTUSH, 1949].

Drawdown at the well face--At the face of the well, $(y - y_1) = 0$, $(x - x_1) = r_w$, and $(x + x_1) = 2x$, where (x_1, y_1) are the coordinates of the well center, and r_w is its radius. For the above values (20) gives, after simplification

$$s_w/(Q/2\pi T) = \sum_{n=1}^{\infty} (1/\delta) [\cos(n\pi r_w/2a) - (-1)^n \cos(n\pi x_1/a)] \left\{ 1 - \operatorname{erfc} [(\pi \delta/4)/\sqrt{u_a}] \right\} \\ \dots \dots \dots (22)$$

where s_w = drawdown at the well face.

The first infinite sum of (22) gives the steady-state drawdown at the well face, which has been tabulated and illustrated graphically [HANTUSH, 1949] for several practical values of the parameters involved, while the second infinite sum may be considered as a superposed transient state. It can be seen that, for all practical values of r_w/a , $[(r_w/a) < 0.001]$ and of u_a , $[u_a > 0.02]$, the transient solution may be considered as independent of the well size, that is, of (r_w/a) if $(r_w/a) < 0.001$.

Figure 2 is prepared to illustrate the variation with time of the drawdown at the face of the well for different values of leakage; the data on which it is based are given in Table 1. The curves shown are constructed for $(r_w/a) = 0.0001$ with the well at the center of the strip.

Figure 4 gives the steady drawdown curves, expressed in terms of the ratio $s_w/(Q/2\pi T)$, for various values of (x_1/a) , (a/B) , and (a/r_w) while Figure 3 gives the transient drawdown, in terms of $s_w/(Q/2\pi T)$ at the well face with the well at the center of the strip and with $(a/r_w) \geq 1000$. The two figures may be used to find the drawdown at the face of a well centered at $(0, 0)$ for values of $(a/r_w) \geq 1000$. Table 2 gives values on which Figure 3 is based.

Table 1--Values of $s_w/(Q/2\pi T)$ in an infinite strip of a leaky aquifer with well at $(0, 0)$ discharging at rate Q ; drawdown is zero at $x = \pm a$; $(a/r_w) = 10,000$

$1/u_a$	a/B		
	0	$\pi/2$	π
0.01	6.619	6.610	6.608
0.02	6.962	6.960	6.944
0.05	7.430	7.409	7.366
0.10	7.770	7.740	7.658
0.20	8.119	8.058	7.886
0.50	8.586	8.433	8.108
1.	8.917	8.653	8.169
2.	9.219	8.781	8.181
5.	9.446	8.817	8.181
10.	9.452	8.818	8.181

Table 2--Values of $s'_w/(Q/2\pi T)$ transient drawdown at well face, with well at center of strip; drawdown is zero at $x = \pm a$; $(a/r_w) \geq 1000$

$1/u_a$	a/B		
	0	$\pi/2$	π
0.01	2.833	2.208	1.573
0.02	2.490	1.858	1.237
0.05	2.022	1.409	0.815
0.10	1.682	1.078	0.523
0.20	1.333	0.760	0.295
0.50	0.866	0.385	0.073
1.	0.535	0.165	0.012
2.	0.233	0.037	0.004
5.	0.0056	0.001	
10.			

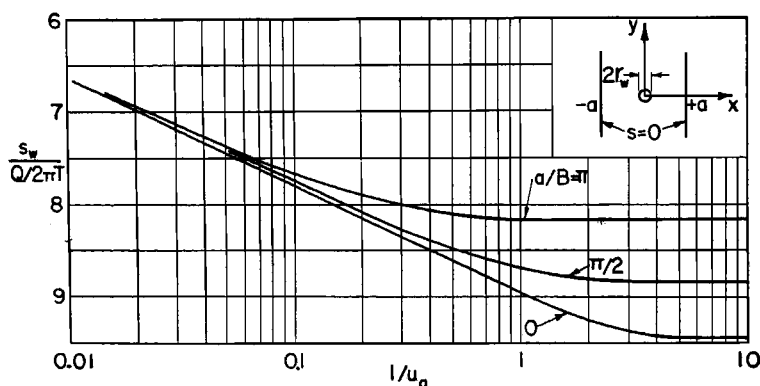


Fig. 2--Time-drawdown curves at well face in an infinite strip of a leaky aquifer with well at $(0, 0)$ and discharging at rate Q ; drawdown is zero at $x = \pm a$; $(a/r_w) = 10,000$

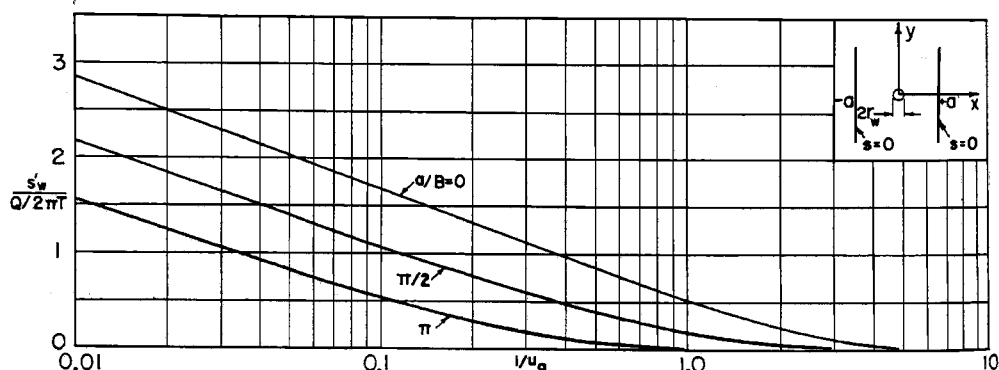


Fig. 3--Transient drawdown curves at well face in an infinite strip of a leaky aquifer with well at $(0, 0)$ and discharging at rate Q ; drawdown is zero at $x = \pm a$; $(a/r_w) \geq 1000$

Drawdown distribution for zero drawdown on the boundaries of a non-leaky aquifer--For a non-leaky aquifer, $B \rightarrow \infty$ or $(1/B) = 0$. Writing zero for $(1/B)$ in (20), the solution becomes

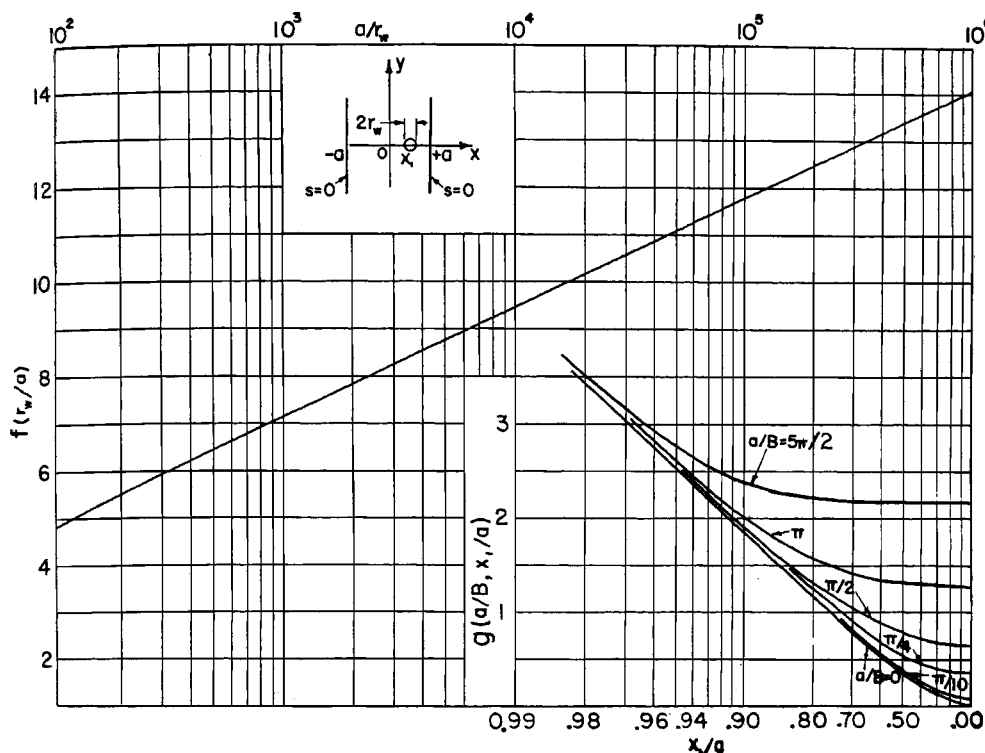


Fig. 4--Steady-state drawdown curves at well face in an infinite strip of a leaky aquifer with well at $(x_1, 0)$ and discharging at rate Q

$$\begin{aligned}
 s_w/(Q/2\pi T) &= \sum_{n=1}^{\infty} (R_n/n) \left\{ \exp[-(n\pi/2a)|y - y_1|] \right\} \\
 &\times \operatorname{erfc}(\sqrt{u_y} - n\pi/4\sqrt{u_a}) - \left\{ \exp[(n\pi/2a)|y - y_1|] \right\} \operatorname{erfc}(\sqrt{u_y} + n\pi/4\sqrt{u_a}) \\
 &\dots\dots\dots (23)
 \end{aligned}$$

The corresponding drawdown at the face of the well is

$$\begin{aligned}
 s_w/(Q/2\pi T) &= \sum_{n=1}^{\infty} (1/n) [\cos(n\pi r_w/2a) - (-1)^n \cos(n\pi x_1/a)] \\
 &\times [1 - \operatorname{erfc}(n\pi/4\sqrt{u_a})] = (1/2) \ln \frac{1 + \cos(\pi x_1/a)}{1 - \cos(\pi r_w/2a)} \\
 &- \sum_{n=1}^{\infty} (1/n) [\cos(n\pi r_w/2a) - (-1)^n \cos(n\pi x_1/a)] \operatorname{erfc}(n\pi/4\sqrt{u_a}) \dots\dots (24)
 \end{aligned}$$

Drawdown distribution for zero flux across the boundaries in a leaky aquifer--The images that result in a vanishing flux across the boundaries are located at the same positions with respect to the boundaries as those in Figure 1; they differ only, but importantly, in the question of sign. All the images have positive signs, that is, they are taken as sinks. The solution is then obtained by adding, instead of subtracting, the two infinite sums of (4). Consequently, in the process of derivation, (13) and (14) are added. The other steps are carried out in the same manner. The solution is finally reduced to

$$\begin{aligned}
s/(Q/2\pi T) = & (\pi B/4a) \left\{ \exp[-(1/B)|y - y_1|] \operatorname{erfc}(\sqrt{u_y} - a/2B\sqrt{u_a}) \right. \\
& - \left. \exp[(1/B)|y - y_1|] \operatorname{erfc}(\sqrt{u_y} + a/2B\sqrt{u_a}) \right\} \\
& + \sum_{n=1}^{\infty} (F_n/\delta) \left\{ \exp[-(\pi\delta/2a)|y - y_1|] \operatorname{erfc}\left\{\sqrt{u_y} - [(\pi\delta/4)/\sqrt{u_a}]\right\} \right. \\
& - \left. \exp[(\pi\delta/2a)|y - y_1|] \operatorname{erfc}\left\{\sqrt{u_y} + [(\pi\delta/4)/\sqrt{u_a}]\right\} \right\} \dots\dots\dots (25)
\end{aligned}$$

where

$$F_n = \cos[n\pi(x - a)/2a] \cos[n\pi(x_1 - a)/2a] \dots\dots\dots (26)$$

Drawdown distribution for zero flux across the boundaries in a non-leaky aquifer--Starting with the equation for an infinite non-leaky aquifer, namely

$$s/(Q/4\pi T) = \int_u^{\infty} e^{-x} (dx/x)$$

and carrying out the steps followed in the development of (25), the drawdown distribution for zero flux across the boundaries of the infinite strip of a non-leaky aquifer is found to be

$$\begin{aligned}
s/(Q/2\pi T) = & (\sqrt{\pi}/2a) |y - y_1| \left\{ [\exp(-u_y)] (1/\sqrt{u_y}) - \sqrt{\pi} \operatorname{erfc}(\sqrt{u_y}) \right\} \\
& + \sum_{n=1}^{\infty} (F_n/n) \left\{ \exp[-(n\pi/2a)|y - y_1|] \operatorname{erfc}(\sqrt{u_y} - n\pi/4\sqrt{u_a}) \right. \\
& - \left. \exp[(n\pi/2a)|y - y_1|] \operatorname{erfc}(\sqrt{u_y} + n\pi/4\sqrt{u_a}) \right\} \dots\dots\dots (27)
\end{aligned}$$

The above equation could have been obtained by taking the limit of (25) (using l'Hopital's rule of limits) as $(1/B) \rightarrow 0$.

It is to be mentioned here that by proper combination of the solutions of the infinite strip, the drawdown distribution can be obtained for the infinite half-strips and for the rectangles along whose boundaries, the flux, the drawdown, or both vanish.

Non-steady Green's function for leaky aquifers--A clear definition of the non-steady Green's function can be given after the following principles are established [SOMMERFELD, 1949, p. 44].

(a) The equation of non-steady plane flow in leaky aquifers is given by (1) and may be written

$$L(s) = (\partial^2 s / \partial x^2) + (\partial^2 s / \partial y^2) - (S/T) (\partial s / \partial t) - (s/B^2) = 0 \dots\dots\dots (28)$$

where $L(s)$ is the differential expression of the equation.

(b) The adjoint equation to (28); if $v(x, y, t)$ is an integrating factor for the equation $L(s) = 0$, then v must satisfy

$$M(v) = (\partial^2 v / \partial x^2) + (\partial^2 v / \partial y^2) + (S/T) (\partial v / \partial t) - (v/B^2) = 0 \dots\dots\dots (29)$$

which is called the adjoint equation of $L(s) = 0$.

(c) The adjoint differential expression; if s and v are any two arbitrary functions of (x, y, t) , then the adjoint $M(v)$ of $L(s)$ is defined by the property that

$$vL(s) - sM(v) = (\partial A / \partial x) + (\partial B / \partial y) + (\partial C / \partial t) \dots\dots\dots (30)$$

where A , B , and C are functions of s , v , and of their derivatives, which make the expression on the left-hand side of the equation integrable.

It can be shown that the above requirement is met if, by virtue of (29), one obtains the expression $M(v)$. Then, by substituting $L(s)$ and $M(v)$ in (30) and equating like terms, one will find that

$$\left. \begin{aligned}
A &= v(\partial s / \partial x) - s(\partial v / \partial x) \\
B &= v(\partial s / \partial y) - s(\partial v / \partial y) \\
C &= -(S/T)sv
\end{aligned} \right\} \dots\dots\dots (31)$$

(d) The delta-function; a function of as many independent variables as may be required, say $s(x, y, z; x_1, y_1, z_1)$ is called a delta function if it vanishes when $Z \rightarrow Z_1$ for all values of $(x, y) \neq (x_1, y_1)$ and it becomes infinite at $(x, y) = (x_1, y_1)$ so that

$$\int_{x_1-\epsilon}^{x_1+\epsilon} \int_{y_1-\epsilon}^{y_1+\epsilon} \delta \, dy \, dx = 1 \quad \dots \dots \dots (32)$$

With these principles in mind, we may now define the non-steady Green's function for leaky aquifers. A function $g(x, y, t; x_1, y_1, \tau)$ is called the non-steady Green's function for leaky aquifers if it has the following properties.

- (a) Is a solution of the adjoint equation of the equation of the non-steady plane flow in leaky aquifers, namely: $M(g) = 0$.
- (b) Is a delta-function.
- (c) Is symmetrical in the two points (x, y) and (x_1, y_1) at any fixed time.
- (d) Either (i) vanishes on the boundary of (ii), its normal derivative, vanishes on the boundary.

If such a function can be found for a given region, and if the initial drawdown is given and the drawdown or the flux are known on the boundary of the region, then the drawdown distribution in the interior of the region will be given by

$$\begin{aligned} s(P, \tau) = & \iint g_0(P, \tau; Q, 0) s_0(Q, 0) \, dA \\ & + (T/S) \int_0^\tau \oint_S \left\{ [g_b(P, \tau; Q_b, t)] [\partial s(Q, t)/\partial n]_b \right. \\ & \left. - s_b(Q_b, t) [\partial g(P, \tau; Q, t)/\partial n]_b \right\} \, dS \, dt \quad \dots \dots \dots (33) \end{aligned}$$

where $s(P, \tau)$ is the drawdown at any point (P) in the region and at any time τ from the start, g_0 and s_0 are the initial Green's function and the initial drawdown respectively; the subscript b indicates evaluation of the functions on the boundary of the region; and n is the outward normal to the elemental length dS . The first integral is taken over all the region, and the second integral is taken over the lateral surface of the three-dimensional space-time prism (or cylinder).

To develop (33), one proceeds as follows: Let Figure 5 represent a space-time prism for which the Green's function is known. Let the point at which the drawdown is sought be (P, τ) and let the general moving point be (Q, t) . The Gauss Theorem (or Green's lemma) in space for the coordinate system of Figure 5 is

$$\iiint [(\partial A/\partial x) + (\partial B/\partial y) + (\partial C/\partial t)] \, dV = \iint [A \cos(n, x) + B \cos(n, y) + C \cos(n, t)] \, d\sigma \quad \dots (34)$$

where A, B, C are functions of (x, y, t) , dV and $d\sigma$ are respectively the elemental volume and the elemental surface of the space considered; and (n, x) , (n, y) , and (n, t) are the angles between the outward normal to the surface and the corresponding axis.

If the volume integrals of the two sides of (30) are taken, one obtains

$$\iiint [vL(s) - sM(v)] \, dV = \iiint [(\partial A/\partial x) + (\partial B/\partial y) + (\partial C/\partial t)] \, dV$$

Letting $v = g$ the Green's function of the space considered, and applying Gauss Theorem, the above equation reduces to

$$\iiint [gL(s) - sM(g)] \, dV = \iint [A \cos(n, x) + B \cos(n, y) + C \cos(n, t)] \, d\sigma$$

The left-hand side of the above equation vanishes because (s) and (g) are respectively solutions of $L(s) = 0$ and $M(g) = 0$; hence the equation becomes

$$\iint [A \cos(n, x) + B \cos(n, y) + C \cos(n, t)] \, d\sigma = 0$$

Table 3--Green's functions for leaky and for non-leaky systems

Shape of region	Green's functions		
	Boundary values	Leaky systems	Non-leaky systems
Infinite plane		$A \cdot (I)$	$B \cdot (I)$
Infinite half-plane with y-axis as the boundary	Vanishes on $x = 0$	$A \cdot (I - II)$	$B \cdot (I - II)$
	Normal derivative vanishes on $x = 0$	$A \cdot (I + II)$	$B \cdot (I + II)$
Infinite quadrant with the boundaries along the coordinate axes	Vanishes on the boundaries	$A \cdot (I - II - III + IV)$	$B \cdot (I - II - III + IV)$
	Vanishes on $x = 0$, and its normal derivative vanishes on $y = 0$	$A \cdot (I + II - III - IV)$	$B \cdot (I + II - III - IV)$
	Normal derivative vanishes on the boundaries	$A \cdot (I + II + III + IV)$	$B \cdot (I + II + III + IV)$
Infinite strip of width equals $(2a)$ with the boundaries parallel to the y-axis	Vanishes on the boundaries	$C \sum_{n=1}^{\infty} R_n \exp(-u_y - D)$	$C \sum_{n=1}^{\infty} R_n \exp(-u_y - E)$
	Vanishes on $x = a$, and its normal derivative vanishes on $x = -a$	$C \sum_{n=1}^{\infty} O_n \exp(-u_y - P)$	$C \sum_{n=1}^{\infty} O_n \exp(-u_y - H)$
	Normal derivative vanishes on the boundaries	$(1/2) C \exp(-u_y - V)$ $- C \sum_{n=1}^{\infty} F_n \exp(-u_y - D)$	$(1/2) C \exp(-u_y)$ $- C \sum_{n=1}^{\infty} F_n \exp(-u_y - E)$

where

$$I = \exp[-r_1^2 S/4T(\tau - t)]$$

$$II = \exp[-r_2^2 S/4T(\tau - t)]$$

$$III = \exp[-r_3^2 S/4T(\tau - t)]$$

$$IV = \exp[-r_4^2 S/4T(\tau - t)]$$

$$A = S/4\pi T(\tau - t) \exp[-T(\tau - t)/SB^2]$$

$$B = S/4\pi T(\tau - t)$$

$$C = \sqrt{\pi} [S/4\pi T(\tau - t) u_a]$$

$$D = \pi^2 [n^2 + (2a/\pi B)^2]/16u_a$$

$$E = (n\pi/4)^2/u_a$$

$$F_n = \cos(n\pi/2a)(x-a) \cos(n\pi/2a)(x_1-a)$$

$$H = [(2n-1)\pi/4]^2/4u_a$$

$$O_n = (-1)^n \sin[(2n-1)\pi/4a](x-a) \cos[(2n-1)\pi/4a]x_1$$

$$P = \pi^2 [(2n-1)^2 + (4a/\pi B)^2]/64u_a$$

$$R_n = \sin(n\pi/2a)(x-a) \sin(n\pi/2a)(x_1-a)$$

$$u_y = (y - y_1)^2 S/4T(\tau - t)$$

$$u_a = a^2 S/4T(\tau - t)$$

$$r_1 = \sqrt{(x_1 - x)^2 + (y_1 - y)^2}$$

$$r_2 = \sqrt{(x_1 + x)^2 + (y_1 - y)^2}$$

$$r_3 = \sqrt{(x_1 - x)^2 + (y_1 + y)^2}$$

$$r_4 = \sqrt{(x_1 + x)^2 + (y_1 + y)^2}$$

$$V = (a/B)^2/u_a$$

Replacing A, B, and C by their values from (31), one obtains

$$\iint \left\{ [g(\partial s / \partial x) - s(\partial g / \partial x)] \cos(n, x) + [g(\partial s / \partial y) - s(\partial g / \partial y)] \cos(n, y) - (S/T) g s \cos(n, t) \right\} d\sigma = 0 \quad (35)$$

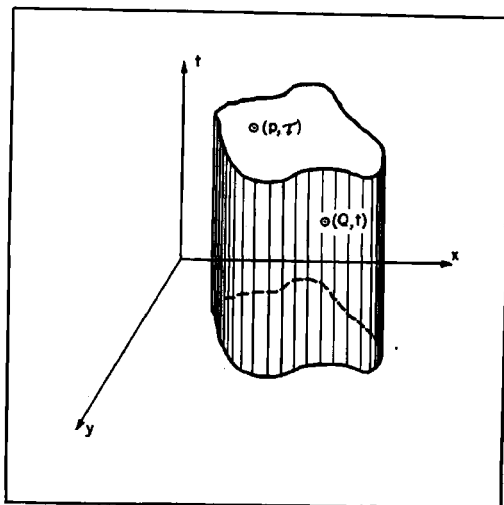


Fig. 5--Closed region for which the Green's function is known

Applying (35) to the surface of the right angled space time prism of Figure 5 excluding the point (P, τ) by a small 2ϵ square (with ϵ chosen arbitrarily small), the total surface integral will break down to the sum of the following surface integrals:

(a) The integral over the base of the prism of Figure 5; that is for $t = 0$. Over the base of the prism, $\cos(n, x) = \cos(n, y) = 0$, $\cos(n, t) = -1$, and $d\sigma = dx dy = dA$. Therefore, the surface integral of (35) becomes

$$(S/T) \iint g_0 s_0 dA$$

where g_0 and s_0 are the value of g and s for $t = 0$.

(b) The integral over the top of the prism where $t = \tau$. Over the top of the prism, $\cos(n, x) = \cos(n, y) = 0$, $\cos(n, t) = 1$, and $d\sigma = dx dy = dA$. Thus the integral of (35) becomes

$$-(S/T) \iint g_\tau s_\tau dA$$

where g_τ and s_τ are the values of g and s for $t = \tau$. But, since g is a delta function, the integral vanishes for all $(x, y) \neq (x_1, y_1)$ except at the point (P, τ) where by the property of the δ -function, it becomes $[-(S/T) s(P, \tau)]$. That is

$$-(S/T) \iint s_\tau g_\tau dA = -(S/T) s(P, \tau) \int_{x_1-\epsilon}^{x_1+\epsilon} \int_{y_1-\epsilon}^{y_1+\epsilon} g_\tau dx dy = -(S/T) s(P, \tau)$$

(c) The integral over the lateral surface of the prism of Figure 5 for $0 < t < \tau$. Over this surface $\cos(n, t) = 0$ and $d\sigma = dS dt$, where dS is the elemental length of the curve S (forming) the boundaries of the top and base of the prism. So the integral of (35) becomes

$$\int_0^\tau \oint_S \left\{ [g_b(\partial s / \partial x)_b - s_b(\partial g / \partial x)_b] \cos(n, x) + [g_b(\partial s / \partial y)_b - s_b(\partial g / \partial y)_b] \cos(n, y) \right\} dS dt$$

which is equal

$$\int_0^\tau \oint_S [g_b(\partial s / \partial n)_b - s_b(\partial g / \partial n)_b] dS dt$$

where the subscript b indicate evaluation of the functions on the boundary, and where n is the outward normal to the curve.

Adding these integrals and rearranging terms, (33) will be obtained. This representation of $s(P, \tau)$ is general since the singular point (P, τ) can be situated at any arbitrary point $x = x_1$, $y = y_1$, and $t = \tau$.

Non-steady Green's functions for the infinite plane and for the infinite strip--The solutions obtained in the foregoing discussion, namely, those given by (2), (20), and (26), are solutions of the differential equation $L(s) = 0$ for the case in which the initial time is zero. They will be solutions for the case of a general initial time, $t = \tau$ if t is replaced by $(t - \tau)$. Furthermore, they fulfill the last two properties of the non-steady Green's function as defined above. To fulfill the first property of the Green's function, that is, to transform them into solutions of the adjoint equation $M(g) = 0$, one needs only to compare the two equations $L(s) = 0$ and $M(g) = 0$. This comparison will indicate that the solutions of $L(s) = 0$, with the sign of $(t - \tau)$ reversed, will be the corresponding solutions of $M(g) = 0$. It can easily be shown that the derivative of $s/(Q/4\pi T)$ in these solutions with respect to time multiplied by $S/4\pi T$ will fulfill the second property of the Green's function. Thus, the derivatives, with respect to time, of the solutions given by (2), (20), and (26) will be the non-steady Green's functions for the region considered if, after performing the differentiation, t is replaced by $(\tau - t)$.

It is of interest to mention here that the previous discussion will also apply to non-leaky systems if one looks upon $L(s)$ and $M(g)$ as the differential expressions representing the differential equations for non-leaky systems. Therefore, the non-steady Green's function for non-leaky systems can also be found (as outlined above) from the corresponding solutions of the non-leaky systems.

It is also to be noticed that while the solutions of $L(s) = 0$ are defined for the future of τ , that is, for $t > \tau$, the solutions of $M(g) = 0$ are defined for the past of τ , that is, for $t < \tau$.

Based upon the above analysis the non-steady Green's functions for several regions are given in Table 3.

Finally it may be mentioned that by proper combination of the Green's functions for the infinite strips, the Green's functions for the infinite half-strips and for the rectangles can be obtained.

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College of Engineering (M. S. H.),
Baghdad, Iraq

Brigham Young University (C. E. J.),
Provo, Utah

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